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Assignment-03

FLAT - Tutorial Sheet - 01 (Unit-3)

- Section - A

1. G is $S \to aS \mid bS \mid \wedge$, find $L(G)$.

Ans
$S \to aS$
$S \to bS$
$S \to \wedge$
$L(G) = \{ w \in (a,b)^* \}$

2. Define a context free grammer.

Ans. A Context free grammer (CFG) Consists of four components : a set of non-terminal symbols, a set of terminal symbol, a start symbol and production rules.

3. What is derivation/Parse Tree.

Ans A derivation or parse tree is a tree structure that represents the way a string is generated from a grammer in formal languages, especially in the Context of context-free grammer.

4. Write the definition of $L(G)$.

Ans for a grammer G, the language of the grammer, denoted by $L(G)$ is the set of all strings that can be generated using the production rules of G, starting from the start symbol.

5. L be the set of all Palindrome over $\{a,b\}$ construct Grammer.

Ans
$S \to aSa$
$S \to bSb$
$S \to a$
$S \to b$
$S \to \wedge$

This grammer generates all Palindrome strings

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over {a, b}, such as ^, a, b, aa, bb, aba, bab, etc.

- Section-B (7×3=21)

1. Consider the production rules of grammar G. prove L = $\{a^n b^m \mid m > n, n >= 0\}$ is generated by G.
$S \to AbB, A \to aAb / ^, B \to bB / ^$.

Ans Given;

$S \to AbB$
$\cancel{S} \to aAb / ^$
$S \to bB / ^$

$S \to AbB$
$S \to aAbbB$
$S \to aaAbbbB$
$S \to aaAbbbbB$
$S \to aabbbb \quad \{A \to ^ \ \& \ B \to ^\}$

$S \to a^n b^m$ and $m > n, n \geq 0$

- A generates strings with same number of a's followed by same number of b's, and B generates a sequence of b's of no symbols

- So, from $S \to AbB$; A can generate strings of the form $a^n b^n$, $n \geq 0$ and $S \to A$ the production in bB in $S \to AbB$ adds at least one more b.
- So this ensures total no. of b's will always be greater than a's.

Hence $L = \{a^n b^m \mid m > n, n \geq 0\}$ is the language generated by G.

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2. if $G = (\{S, C\}, \{a, b\}, \{S \to aCa, C \to aCa, C \to b\}, S)$.
Find $L(G)$.

Ans Non-terminal symbols: $\{S, C\}$
Terminal symbols: $\{a, b\}$
Production rules:
$S \to aCa$
$C \to aCa$
$C \to b$

Start Symbol: S
$S \to aCa$
$S \to aaCaa$
$S \to aabaa$ or $S \to aaaCaaa$
$\to aaabaaa$

General structure of strings in $L(G)$:
- All strings in $L(G)$ are of the form $a^n b a^n$, where $n \geq 1$. The no. of a's before and after the b is same.

Thus the language $L(G)$ generated by the grammer is
$$L(G) = \{ a^n b a^n \mid n \geq 1 \}$$

3. if $L(G) = (S \to a^i b^i c^j,\ i >= 1, j >= 0)$, Find Grammer
Ans $\Sigma = (a, b, c)$
ab, abc, aabbcc
$S \to SC$, $S \to A$, $A \to aAb$, $A \to ab$

$S \to abSC$
$S \to ab$
$S \to \wedge$

Grammer that generates the Language

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5). $L(G) = \{ a^i b^i c^j \mid i \geq 1, j \geq 0 \}$ is

$S \rightarrow abSc \mid ab$.

- Section-C (11 * 3 = 33)

1. Explain the Pumping Lemma for Regular Language. And Using pumping Lemma Prove that the Language $A = \{ yy \mid y \in (0,1)^* \}$ is not regular.

Ans If A is a regular language, then A has a pumping length 'P' such that any string 's' where $|s| \geq P$ may be divided into 3 parts $s = xyz$ such that the following conditions must be true:

i) $xy^iz \in A \quad \forall \quad i \geq 0$

ii) $|y| > 0$

iii) $|xy| \leq P$

- $A = \{ yy \mid y \in \{0,1\}^* \}$

To prove : A is not regular

proof :-

first half of string is same as second half.

A is regular

pumping length $= P = 6$

$S = 0^P 1 0^P 1$ then $S = 00000010000001$

$x\ y\ z$

CASE I :- $\underbrace{00}_{x}\underbrace{0000}_{y}\underbrace{10000001}_{z}$

CASE II :-

$xy^iz \Rightarrow xy^3z$ for $P = 3$

0000000001000000 1

∴ I half ≠ II half ∉ A

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$|y| > 0$ correct
$|xy| \le p$ correct $6 \le 6$

S can not be pumped
So it is not regular.

2. Explain the Normal form and convert the following CFG in
$G = (\{S, A, B\}, \{a, b\}, \{S \to aAbB, A \to aA \mid a, B \to bB \mid b.\}, S)$ in CNF
$G = (\{S\}, \{a, b\}, \{S \to absb \mid aa\}, S)$ in GNF.

Ans Normal forms refer to specific standardized forms of CFG (context free Grammar).

- Chomsky Normal form (CNF)
A CFG is in a CNF if all of its production rules satisfy one of the following conditions:

(i) $A \to BC$ where $(A, B, C) \in V_n$ & $(B, C) \notin S$
(ii) $A \to a$; $a \in \Sigma$
(iii) Grammar may have production $S \to \wedge$ if S is start symbol & $\wedge$ is empty string.

- Greibach Normal Form (GNF):
Rules -
i) $A \to a\alpha$, $A \in V_n$ & α is a string of non-terminals & $a \in \Sigma$.

(a)

$S \to aAbB$
$A \to aA$
$A \to a$] CNF
$B \to bB$
$B \to b$) CNF

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- $S \to aAbB$
  Let $X \to a$, $Y \to b$
  then $S \to XAYB$

- $A \to aA$
  Let $Z \to a$
  So, $A \to ZA$

- $B \to bB$
  Let $W \to b$
  So $B \to WB$

Now we have following CNF:

$S \to XAYB$
$A \to ZA$
$A \to Z$
$B \to WB$
$B \to W$
$X \to a$
$Y \to b$
$Z \to a$
$W \to b$

This is the CFG in Chomsky Normal Form.

(b) CFG: $G = (\{S\}, \{a, b\}, \{S \to absb \mid aa\}, S)$ to GNF.

$S \to absb$
$S \to aaS'$
$S' \to bS' \mid \Lambda$
S' is a new non-terminal.

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Productions are :-

$S \to aaS'$

$S' \to bS'$

$S' \to \wedge$

Each of these are in GNF thus it is the CFG in Greibach Normal form.

Q3 (A) Explain the chomsky classification of Grammars in details with example

(B) Consider the grammer $P = \{S \to aS \mid aSbS \mid \epsilon\}$ is ambiguous by constructing:

(i) two parse trees

(ii) two Leftmost derivation

Ans (A) A chomsky classification of grammer divide grammars into 4 types i known as Chomsky Hierarchy :-

1. Type 0
2. Type 1
3. Type 2
4. Type 3

- Type 0 : (Recursively Enumerable languages) :- it can be generate any language that can be recognized by turing machine, and has no restrictions on its production rules.

- Type 1 :- (Context ~~Structure~~ Sensitive language) :- Left hand side can be a string of V_n & Σ and its length must be less than or equal to that of string on RHS.

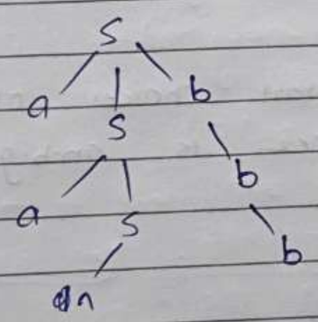
- Type 2 (Context Free Language)
  it generates languages which can be parsed down by a PDA (Push Down Automata).

- Type 3 (Regular languages)
  These can be recognized by finite automata.

(B) Let $P = \{ S \to aS \mid aSbS \mid \wedge \}$

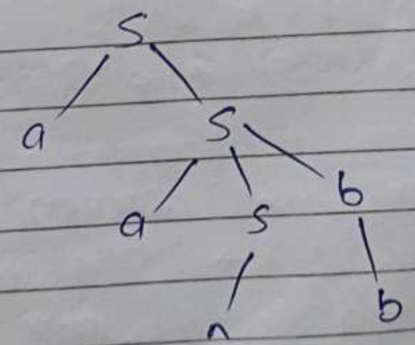
i) Two parse trees

- First we will use production $S \to aSbS$ twice



$S \to aSbS \to aaSbSb \to aabSbSb \to aababb$

- Second parse tree using production $S \to aS$ then $S \to aSbS$.



$S \to aS \to aaSbS \to aabSbS \to aababb$.

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Grammar is ambiguous as the string "aababb" can be derived in two different ways.

(ii) Using two left most derivations.

1. $S \to aSbS$
$\to aaSbSb$
$\to aabSbSb$
$\to aababb$

2. $S \to aS$
$\to aaSbS$
$\to aabSbS$
$\to aababb$

there are different choices of Productions hence grammar is ambigous.

x x x x

The end